CHE-311

Hw set 5

6A2a) Elevation change

$\Delta z = 50 \sin 45^\circ = 35.36 ft$

Reynold number

$Re = \frac{\rho V D}{\mu} \approx 1.85 \cdot 10^4$

$f \approx 0.028$

Major head loss, from pipe friction

$h_f = f \frac{L V^2}{D 2g} = (0.028)\left(\frac{95}{0.2557}\right)\left(\frac{0.781^2}{2(32.2)}\right) \approx 0.098 ft$

Minor Loss $K_{45} \approx 0.35$ $K_{total} = 2(0.35) = 0.7$

$h_m = K \frac{V^2}{2g} = 0.7 \frac{0.781^2}{2(32.2)} \approx 0.0066 ft$

Flow velocity in the pipe

$Q = 18\left(\frac{0.13368 ft^3}{1 gal}\right)\left(\frac{1}{60}\right) = 0.04010 \frac{ft^3}{s}$

$A = \frac{\pi D^2}{4} = \frac{\pi (0.25778)^2}{4} = 0.05134 ft^2$

$V = \frac{Q}{A} = \frac{0.04010}{0.05134} = 0.781 \frac{ft}{s}$

Energy Equation

$\frac{P_1}{\gamma} = \Delta z + h_f + h_m = 35.36 + 0.098 + 0.0066 = 35.46 ft$ of water

$P_1 = \gamma h$

$\gamma = 62.3 \frac{lbf}{ft^3}$ $P_1 = \frac{62.3 (35.46)}{144} \approx 15.3 psi$

6A2b) $\% = \frac{h_f}{h_{total}} * 100 = \frac{0.098}{35.46} 100 = 0.28\%$

6A3a) Method A $\Delta P = 2 f \rho V^2 \frac{L}{D}$

$V = \sqrt{\frac{\Delta P D}{2 f \rho L}} = \sqrt{\frac{(1723.7)(0.1524)}{2(0.006)(998.2)(402.34)}} \approx 0.234 \frac{m}{s}$

$Re = \frac{\rho V D}{\mu} = \frac{(998.2)(0.234)(0.1524)}{1.002 \cdot 10^{-3}} \approx 3.56 \cdot 10^4$

$f = 0.0741 Re^{-0.25} = 0.0741 (3.56 \cdot 10^4)^{-0.25} = 0.00562$

using $f = 0.00562$

$V = \sqrt{\frac{(1723.7)(0.1524)}{2(0.00562)(998.2)(402.34)}} \approx 0.242 \frac{m}{s}$ The Solution is converged

$Re = \frac{(998.2)(0.242)(0.1524)}{1.002 \cdot 10^{-3}} \approx 3.67 \cdot 10^4$ $V \approx 0.242 \frac{m}{s}$

$f = 0.0741 (3.67 \cdot 10^4)^{-0.25} \approx 0.00561$

Volumetric flow rate

$A = \frac{\pi D^2}{4} = \frac{\pi (0.1524)^2}{4} = 0.01824 m^2$ $Q = AV = (0.01824)(0.242) = 0.00441 \frac{m^3}{s}$

$Q = 0.00441 \cdot 264.172 \cdot 3600 \approx \boxed{4.19 \cdot 10^3 \frac{gal}{hr}}$

Method B

Sub for f:

$\Delta P = 2 f \rho V^2 \frac{L}{D}$ $f = 0.0791 \left(\frac{\rho V D}{\mu}\right)^{-0.25}$ $\Delta P = 2(0.0791)\left(\frac{\rho V D}{\mu}\right)^{-0.25} \rho V^2 \frac{L}{D}$

$V^{1.75} = \frac{\Delta P D}{0.158 \rho L}\left(\frac{\rho D}{\mu}\right)^{0.25} = \frac{(172.37)(0.1524)}{0.158(998.2)(402.34)}\left(\frac{(998.2)(0.1524)}{1.002 \cdot 10^{-3}}\right)^{0.25}$

$V = 0.234 \frac{m}{s}$

$Q = AV = (0.01824)(0.234) = 0.00436 \frac{m^3}{s} = \boxed{4.14 \cdot 10^3 \frac{gal}{hr}}$

6A9) $\frac{dP}{dz} = 150 \frac{(1-\varepsilon)^2}{\varepsilon^3 D_p^2} \mu u + 1.75 \frac{(1-\varepsilon)}{\varepsilon^3 D_p} \rho u^2$ $G = \rho u$ $u = \frac{G}{\rho}$

Sub into Ergun eqn

$-\frac{dP}{dz} = \frac{150(1-\varepsilon)^2}{\varepsilon^3 D_p^2} \mu \frac{G}{\rho} + 1.75 \frac{(1-\varepsilon)}{\varepsilon^3 D_p} \frac{G^2}{\rho}$ $\rho = \frac{MP}{RT}$

$\frac{P_1^2 - P_2^2}{2L} = \frac{RT}{M}\left[150 \frac{(1-\varepsilon)^2}{\varepsilon^3 D_p^2} \mu G + 1.75 \frac{(1-\varepsilon)}{\varepsilon^3 D_p} G^2\right]$ $P_1 = 2.533 \cdot 10^6 Pa$ $P_2 = 3.040 \cdot 10^5 Pa$

$A = 150 \frac{(1-\varepsilon)^2}{\varepsilon^3 D_p^2} = 150 \frac{(0.59)^2}{(0.41)^3 (0.0015875)^2} = 3.006 \cdot 10^8$

$B = 1.75 \frac{(1-\varepsilon)}{\varepsilon^3 D_p} = 1.75 \frac{0.59}{(0.41)^3 (0.0015875)} = 9437 \cdot 10^3$

$\frac{P_1^2 - P_2^2}{2L} \frac{M}{RT} = 3.328 \cdot 10^7$ $A\mu G + B G^2 = 3.328 \cdot 10^7$

$(3.006 \cdot 10^8)(1.495 \cdot 10^{-5}) G + (9.437 \cdot 10^3) G^2 = 3.328 \cdot 10^7 = 59.15 \frac{kg}{m^2 s}$

Convert to mass flow rate

$A_c = \frac{\pi D^2}{4} = \frac{\pi (0.1016)^2}{4} = 0.08107 m^2$ $\dot{m} = G A_c$

$= (59.15)(0.08107) = \boxed{0.479 \frac{kg}{s}}$

6B2a) Reynolds number Friction factor definition

$R_L = \frac{\rho V_\infty L}{\mu}$ $f = \frac{F_k}{\frac{1}{2}\rho V_\infty^2 A}$ $A = 2WL$ $f = \frac{F_k}{\frac{1}{2}\rho V_\infty^2 (2WL)}$

6B2b) Turbulent Flow

$F_k = 0.74 \rho V_\infty^2 WL R_L^{-\frac{1}{5}}$ $f = \frac{F_k}{\frac{1}{2}\rho V_\infty^2 (2WL)} = \frac{F_k}{\rho V_\infty^2 WL} = \frac{0.74 \rho V_\infty^2 WL R_e^{-\frac{1}{5}}}{\rho V_\infty^2 WL}$

$f = 0.74 R_L^{-\frac{1}{5}}$

6B3) $v_z(x) = v_{max}\left(1 - \frac{x^2}{B^2}\right)$

Mean velocity

$\langle v_z \rangle = \frac{1}{2B}\int_{-B}^{B} v_z(x)dx$; $\langle v_z \rangle \frac{v_{max}}{2B}\int_{-B}^{B}\left(1-\frac{x^2}{B^2}\right)dx$; $\langle v_z \rangle = \frac{v_{max}}{2B}\left(2B - \frac{2B}{3}\right) = \frac{2}{3}v_{max}$

Wall shear stress differentiate $x = B$

$\tau_w = \mu \left|\frac{dv_z}{dx}\right|_{x=B}$ $\frac{dv_z}{dx} = -\frac{2v_{max}x}{B^2}$ $\left.\frac{dv_z}{dx}\right|_{x=B} = -\frac{2v_{max}}{B}$ $\tau_w = \mu\frac{2v_{max}}{B}$

$\tau_w = \mu\frac{2}{B}\left(\frac{3}{2}\langle v_z \rangle\right) = \frac{3\mu\langle v_z \rangle}{B}$

$f = \frac{\tau_w}{\frac{1}{2}\rho\langle v_z \rangle^2} = \frac{\frac{3\mu\langle v_z \rangle}{B}}{\frac{1}{2}\rho\langle v_z \rangle^2} = \frac{6\mu}{B\rho\langle v_z \rangle}$ $R_e = \frac{2B\langle v_z \rangle \rho}{\mu}$

$\frac{12}{R_e} = \frac{12\mu}{2B\langle v_z \rangle \rho} = \frac{6\mu}{B\rho\langle v_z \rangle}$ $f = \frac{12}{R_e}$

$A = 2BW$ $P = 2W$ $D_h = 4R_h = 4B$ $R_h = \frac{A}{P} = B$

$f = \frac{16}{R_{eh}}$ $R_{eh} = \frac{D_h\langle v_z \rangle \rho}{\mu} = \frac{4B\langle v_z \rangle \rho}{\mu}$

$f = \frac{16\mu}{4B\langle v_z \rangle \rho} = \frac{4\mu}{B\langle v_z \rangle \rho}$ $\frac{f_{hydraulic}}{f_{exact}} = \frac{4}{6} = \frac{2}{3}$